SOLVATHON 2026 – Problem Statement 3

Bus Ticket Data Analytics & Intelligent Dashboard

The Challenge

Can you transform raw bus ticket collection data into an intelligent dashboard that reveals meaningful insights and helps improve transport planning and decision-making?

Transforming daily bus ticket data into meaningful insights for better transport planning and decision-making.

1. Problem Overview

The Bus Department maintains daily ticket collection data containing information about bus travel and ticket transactions. This data can provide valuable insights into passenger travel patterns, route demand, revenue and operational trends.

However, analysing large amounts of raw data manually can be difficult. The department needs a simple way to understand this information and use it for better decision-making.

The challenge is to develop an intelligent data analytics dashboard that converts the available bus ticket data into meaningful insights through analysis, visualisations and predictions wherever possible.

2. What Students Need to Build

Develop a comprehensive, interactive dashboard using the actual data file provided by the Bus Department.

Students should:

- Understand the dataset, its columns, records and available information.
- Identify useful patterns, relationships and trends.
- Analyse ticket collection, route-wise demand and travel patterns wherever possible.
- Present findings through interactive charts, graphs and tables.
- Explore predictions of future demand and revenue if sufficient historical data is available.

3. Suggested Areas of Analysis

The following are examples to help students get started:

Analysis Area	**Possible Exploration**
Ticket & Passenger Analysis	Ticket transactions and passenger travel patterns
Revenue Analysis	Total collection, route-wise revenue and trends
Route Analysis	High-demand and low-demand routes, popular destinations
Time-Based Analysis	Daily, monthly, weekday and weekend trends
Operational Analysis	Ticket collector-wise collection and transaction summaries
Predictive Analysis	Future passenger demand and revenue forecasts

Note: These are only illustrative examples. Students must thoroughly explore the dataset and identify as many meaningful analysis opportunities, insights and outcomes as the available data supports. Do not limit your solution to the examples listed above.

4. Expected Dashboard

The dashboard should be simple, interactive and easy for department officials to understand. It may include:

- Overall summary of ticket transactions and recorded collection.
- Route-wise and destination-wise analysis.
- Passenger travel and demand patterns.
- Revenue and ticket collection trends.
- Date-wise comparisons and interactive filters.
- Predictive insights wherever feasible.

The final dashboard should be based on the actual dataset, not assumed information.

5. Important Instructions

- The actual Bus Department data file will be provided to participating teams for analysis.
- Understand the dataset before deciding what to build.
- Ensure all calculations and visualisations accurately represent the original data.
- Clearly distinguish actual values from predictions.
- Do not assume ticket transactions represent unique passengers unless the data supports it.
- Protect any personal or sensitive information in the dataset.

6. Expected Deliverables

1. Dataset Understanding: Identify the available fields and data structure.
2. Data Analysis: Clean the data and identify meaningful insights.
3. Interactive Dashboard: Develop a functional dashboard with relevant visualisations and filters.
4. Predictive Insights: Include forecasting wherever the data supports it.
5. Working Demonstration: Demonstrate the solution using the actual dataset.

7. Evaluation Criteria

Project Readiness Level

- Understanding and analysis of the actual dataset.
- Functionality and interactivity of the dashboard.
- Accuracy and usefulness of visualisations.
- Working product demonstration.

Feasibility of Implementation

- Ease of use for department officials.
- Practical value of the insights generated.
- Compatibility with the department's actual data.
- Ability to update the dashboard with new data.

Don't just display the data. Understand it, analyse it and turn it into actionable insights.